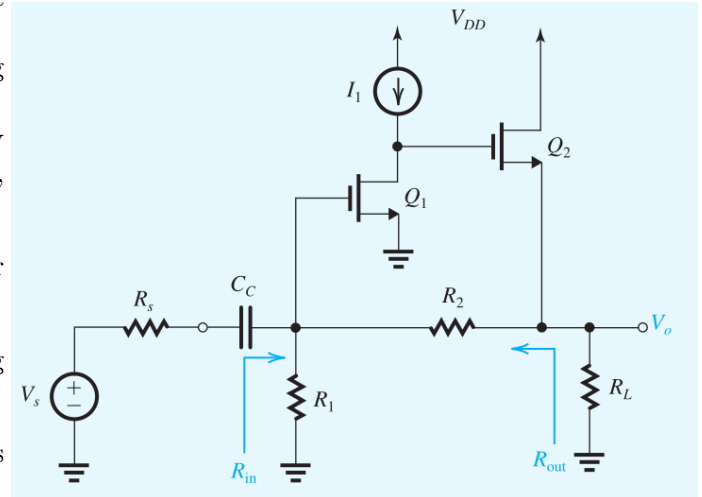


Question 3

Figure below shows a shunt–shunt feedback amplifier. The MOSFETs have $V_{tn} = 0.6$ V, $V_A = 20$ V, and $\mu_n C_{ox} = 200 \mu\text{A}/\text{V}^2$. The power supply $V_{DD} = 3.3$ V, and $R_L = 2$ k Ω . The coupling capacitor C_c can be assumed to be very large.

- Perform a DC design to meet the following specifications: $I_{D1} = 100 \mu\text{A}$, $I_{D2} = 1$ mA, $I_{R2, R1} = 10 \mu\text{A}$, $V_{OV1} = V_{OV2} = 0.2$ V. Neglect the Early effect. Specify the values required for I_1 , R_1 , R_2 , $(W/L)_1$, and $(W/L)_2$.
- Find an expression for β and hence an expression for the ideal value of V_o/V_s .
- Find the value of R_s that results in V_o/V_s being ideally -6 V/V.
- Find the A circuit and use it to determine the values of A , R_{in} , and R_{out} .
- Find the value obtained for V_o/V_s .
- Find R_{in} and R_{out} .



Solⁿ: $I_{D1} = I_1 = 100 \mu\text{A}$ $\therefore I_1 = 100 \mu\text{A}$

$$I_{D1} = \frac{1}{2} (\mu_n C_{ox}) \left(\frac{W}{L}\right)_1 V_{OV1}^2 \quad \therefore \left(\frac{W}{L}\right)_1 = \frac{2I_D}{(\mu_n C_{ox}) V_{OV1}^2}$$

$$\therefore \left(\frac{W}{L}\right)_1 = 25 \quad = \frac{2 \times 100 \times 10^{-6}}{200 \times 10^{-6} \times (0.2)^2} = 25$$

$$I_{D2} = 1 \text{ mA} = I_{S2} = I_{R_L} + I_{R2} = I_{R_L} + 10 \mu\text{A}$$

$$I_{R_L} = 10^{-3} - (10 \times 10^{-6}) = 0.99 \text{ mA}$$

$$V_o = V_s = I_{R_L} R_L = 0.99 \times 10^{-3} \times 2 \times 10^3 = 1.98 \text{ V}$$

$$V_{GS1} = V_{OV1} + V_{tn} = 0.2 + 0.6 = 0.8 \text{ V} \quad \therefore V_{S1} = 0 \quad \therefore V_{G1} = 0.8 \text{ V}$$

$$R_2 = \frac{V_{S2} - V_{G1}}{I_{R2}} = \frac{1.98 - 0.8}{10 \times 10^{-6}} = \underline{\underline{118 \text{ k}\Omega}}$$

C_c is blocking the DC current. So $I_{R_1} = I_{R_2} = 10 \mu A$

$$R_1 = \frac{V_{G1}}{I_{R_1}} = \frac{0.8}{10 \times 10^{-6}} = \underline{\underline{80 \text{ k}\Omega}}$$

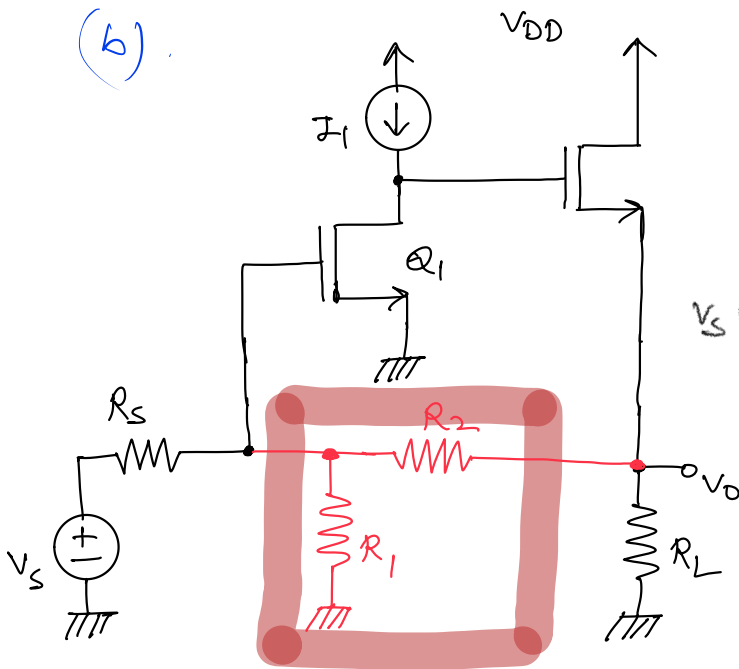
$$I_{D_2} = \frac{1}{2} (\mu_n C_{ox}) \left(\frac{W}{L}\right)_2 V_{ov_2}^2 \Rightarrow \left(\frac{W}{L}\right)_2 = \frac{2 I_{D_2}}{(\mu_n C_{ox}) V_{ov_2}^2}$$

$$\underline{\underline{\left(\frac{W}{L}\right)_2 = 250}}$$

$$= \frac{2 \times 1 \times 10^{-3}}{200 \times 10^{-6} \times (0.2)^2}$$

$$= 250$$

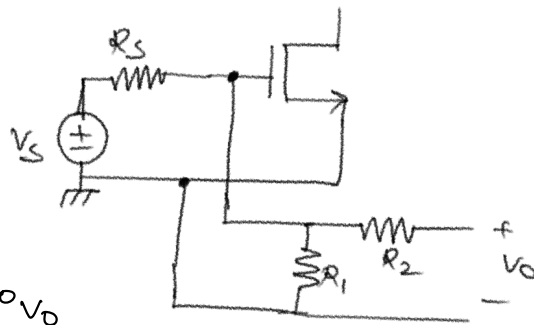
(b)



at O/p side Entire voltage we are picking.

at I/p side ??

→ dealing with current



Transresistance Amp.



$$I_1 = G_1 V_1 + - V_2$$

$$I_2 = \beta V_1 + G_2 V_2$$

$$\beta = \frac{I_2}{V_1} \Big|_{V_2=0}$$

$$G_1 = \frac{I_1}{V_1} \Big|_{V_2=0}$$

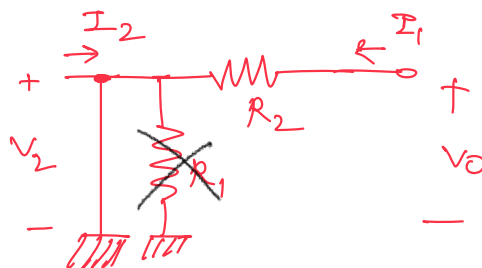
$$G_2 = \frac{I_2}{V_2} \Big|_{V_1=0}$$

$$\frac{0 - V_0}{I_2} = R_2$$

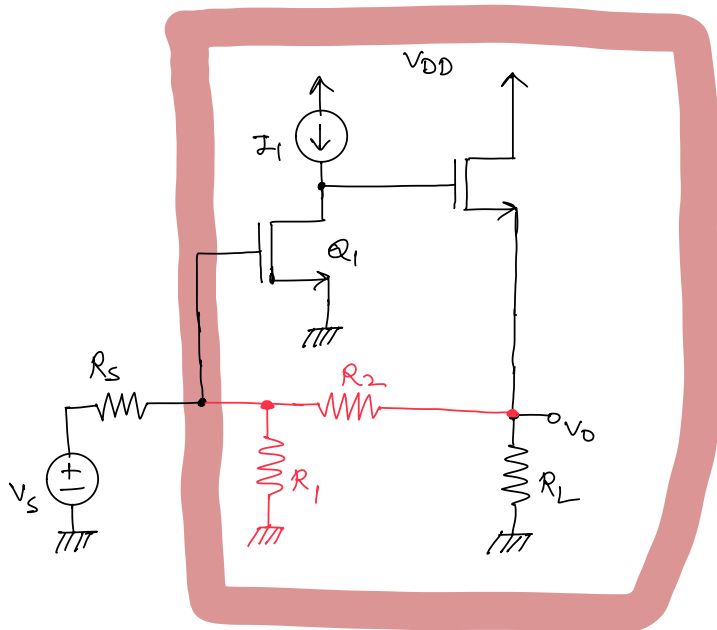
$$\frac{V_0}{I_2} = -R_2$$

$$\frac{1}{\beta} = -R_2$$

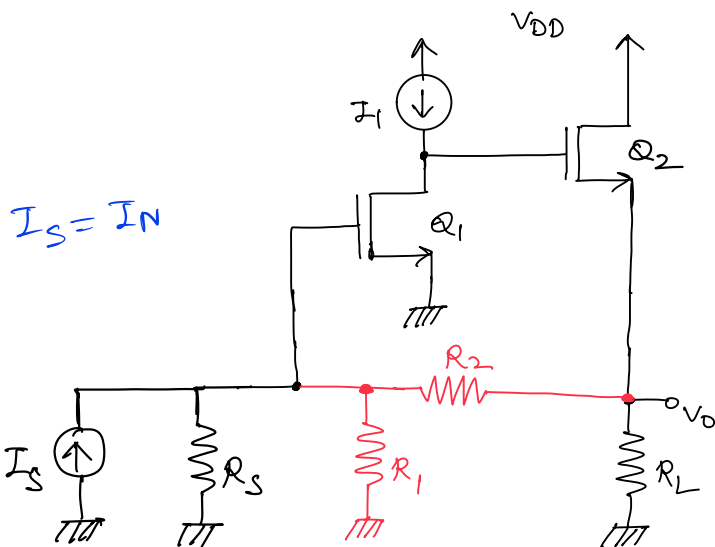
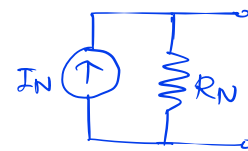
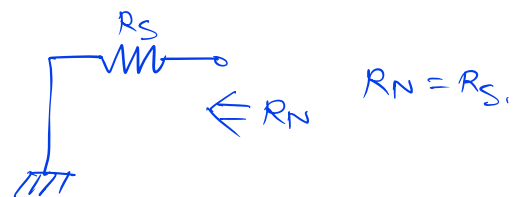
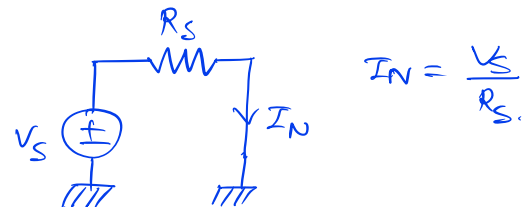
$$\boxed{\beta = -\frac{1}{R_2}}$$



This is a transresistance amp. So, we know $A_{f/ideal} = \frac{1}{\beta}$ when o/p is voltage and input is current. But for this Fig. I/p side is given as voltage source. So we need to convert voltage to current.



Find Norton equivalent circuit to convert the input side to the current.



$$A_{f/ideal} = \frac{1}{\beta}$$

$$\frac{V_O}{I_S} = \frac{1}{\beta}$$

$$\frac{V_O}{V_S/R_S} = -R_2$$

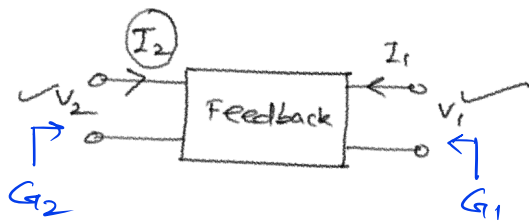
$$\boxed{\left. \frac{V_O}{V_S} \right|_{ideal} = -\frac{R_2}{R_S}} \quad \frac{V \cdot V \cdot I}{= \frac{1}{\beta_V}}$$

$$(C) \quad \left. \frac{V_O}{V_S} \right|_{ideal} = -\frac{R_2}{R_S}$$

$$-6 = -\frac{118k\Omega}{R_S}$$

$$\therefore \boxed{R_S = 19.67 k\Omega}$$

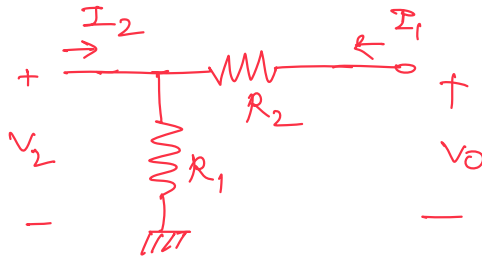
(d).



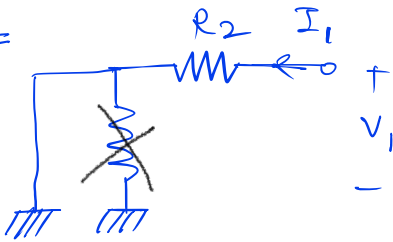
$$I_1 = G_1 V_1 + \dots V_2$$

$$I_2 = \beta V_1 + G_2 V_2$$

$$G_1 = \left. \frac{I_1}{V_1} \right|_{V_2=0} \quad G_2 = \left. \frac{I_2}{V_2} \right|_{V_1=0}$$

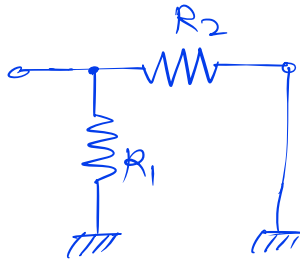


G_1

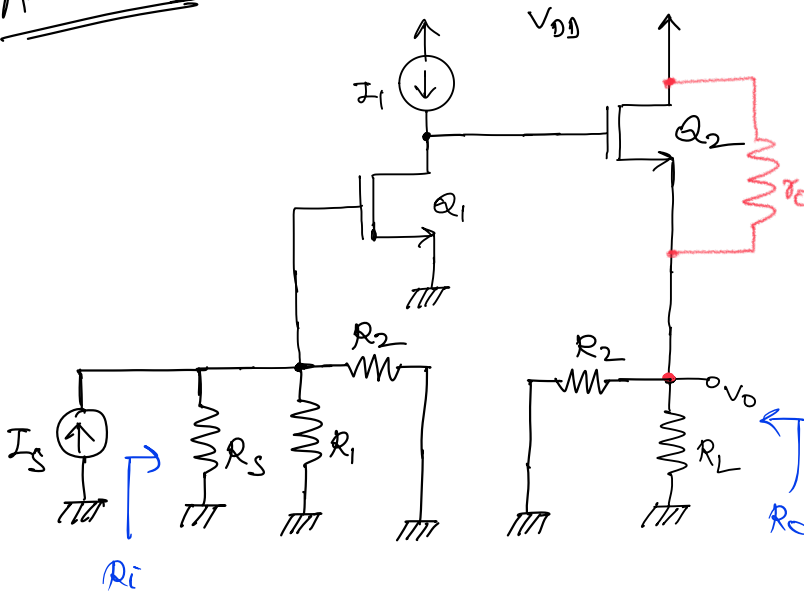


$$\left. \frac{V_1}{I_1} \right|_{V_2=0} = R_2$$

G_2



A-circuit



$Q_1 \text{ CS}$

$$V_{GS1} = I_S (R_S \parallel R_1 \parallel R_2)$$

$$V_{d1} = -g_{m1} V_{GS1} r_{o1}$$

$$= -g_{m1} \frac{V_S}{R_S} (R_S \parallel R_1 \parallel R_2) r_{o1}$$

$Q_2 \rightarrow$ Source follower

$$V_O = \frac{(r_{o2} \parallel R_L \parallel R_2) \cdot V_{d1}}{\frac{1}{g_{m2}} + (r_{o2} \parallel R_L \parallel R_2)}$$

$$V_O = \frac{(r_{o2} \parallel R_L \parallel R_2)}{\frac{1}{g_{m2}} + (r_{o2} \parallel R_L \parallel R_2)} \cdot \left[-g_{m1} \frac{r_{o1}}{R_S} (R_S \parallel R_1 \parallel R_2) V_S \right]$$

$$\frac{V_o}{V_s} = - \frac{g_{m1} r_{o1} (R_S || R_1 || R_2) (R_L || R_2 || r_{o2})}{\left[\frac{1}{g_{m2}} + (r_{o2} || R_L || R_2) \right] R_S} = A.$$

$$g_{m1} = \frac{2I_{D1}}{V_{ov1}} = \frac{2 \times 100 \times 10^{-6}}{0.2} = 1 \text{ mA/V.}$$

$$g_{m2} = \frac{2I_{D2}}{V_{ov2}} = \frac{2 \times 1 \times 10^{-3}}{0.2} = 10 \text{ mA/V.}$$

$$r_{o1} = \frac{V_A}{I_{D1}} = \frac{20}{100 \times 10^{-6}} = 200 \text{ k}\Omega$$

$$r_{o2} = \frac{V_A}{I_{D2}} = \frac{20}{10^{-3}} = 20 \text{ k}\Omega.$$

$$A = - \frac{10^{-3} \times 200 \times 10^3 \times (19.67 \text{ k} || 80 \text{ k} || 118 \text{ k}) (2 \text{ k} || 80 \text{ k} || 20 \text{ k})}{19.67 \times 10^3 \times [100 + (20 \text{ k} || 2 \text{ k} || 118 \text{ k})]}$$

$$= - \frac{200 \times 13.93 \times 10^3 \times 1.78 \times 10^3}{19.67 \times 10^3 \times [100 + 1.79 \times 10^3]} = - \frac{4.96 \times 10^3 \times 10^3}{19.67 \times 1.89 \times 10^3}$$

$$= - \frac{4.96 \times 10^3}{19.67 \times 1.89} = -133.41 \text{ V/V.}$$

$$R_i = (R_S || R_1 || R_2) = 19.67 \text{ k} || 80 \text{ k} || 118 \text{ k} = 13.93 \text{ k}\Omega.$$

$$R_o = (R_L || R_2 || r_{o2}) = 2 \text{ k} || 118 \text{ k} || 20 \text{ k} = 1.79 \text{ k}\Omega.$$

$$(e). \frac{V_o}{V_s} = A_f = \frac{A}{1 + A\beta_v}$$

$$A\beta = A \times \left(-\frac{R_S}{R_2} \right)$$

$$= (-133.41) \times \left(-\frac{1}{6} \right) = 22.24$$

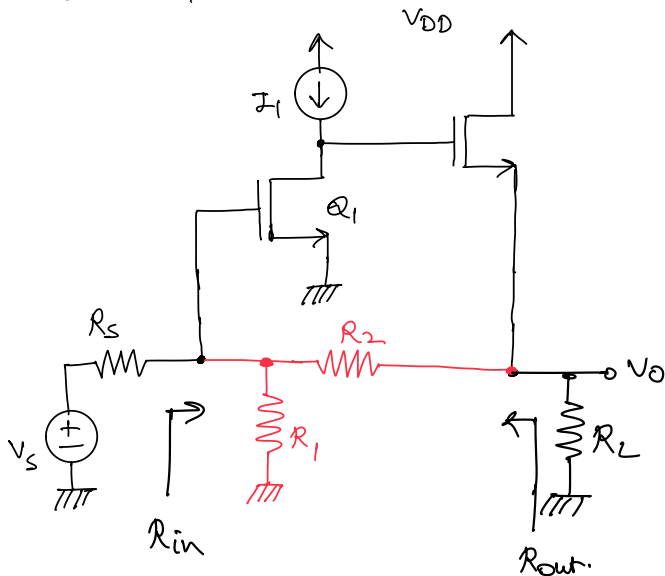
Why not to use $\beta = -\frac{1}{R_2}$.

$A\beta$ dimension becomes $\frac{\text{V}}{\text{V}} \cdot \frac{\text{A}}{\text{V}}$
 $= \text{A/V.}$

But $A\beta$ should be dimensionless.

$$A_f = \frac{-133.41}{1 + 22.24} = -5.74 \text{ V/V.}$$

(f). original



$$R_{if} = \frac{R_i}{1 + A\beta} = \frac{13.93 \text{ k}}{1 + 22.24} = 599.39 \Omega.$$

$$\frac{1}{R_{if}} = \frac{1}{R_s} + \frac{1}{R_{in}}.$$

$$\frac{1}{R_{in}} = \frac{1}{R_{if}} - \frac{1}{R_s}$$

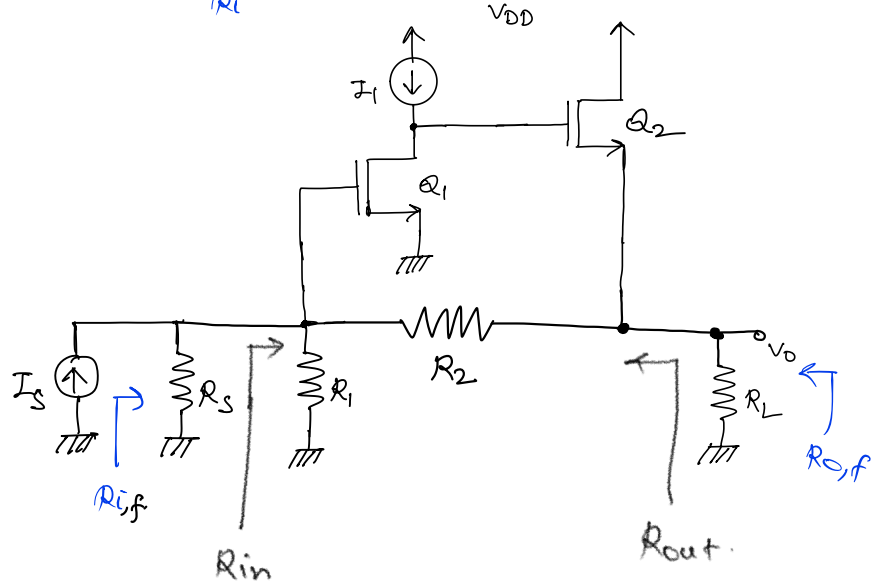
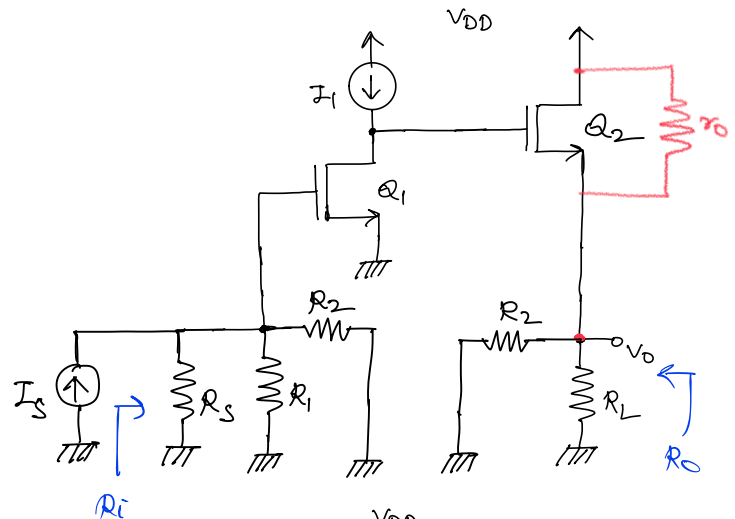
$$R_{in} = \frac{1}{\frac{1}{R_{if}} - \frac{1}{R_s}} = \frac{1}{\frac{1}{599.39} - \frac{1}{19.67 \times 10^3}} = 618.23 \Omega.$$

$$R_{o,f} = \frac{R_o}{1 + A\beta} = \frac{1.79 \text{ k}}{1 + 22.24} = 77.02 \Omega.$$

$$\frac{1}{R_{o,f}} = \frac{1}{R_L} + \frac{1}{R_{out}} \Rightarrow \frac{1}{R_{out}} = \frac{1}{R_{o,f}} - \frac{1}{R_L}$$

$$R_{out} = \frac{1}{\frac{1}{R_{o,f}} - \frac{1}{R_L}} = \frac{1}{\frac{1}{77.02} - \frac{1}{2 \times 10^3}} = 80.10 \Omega.$$

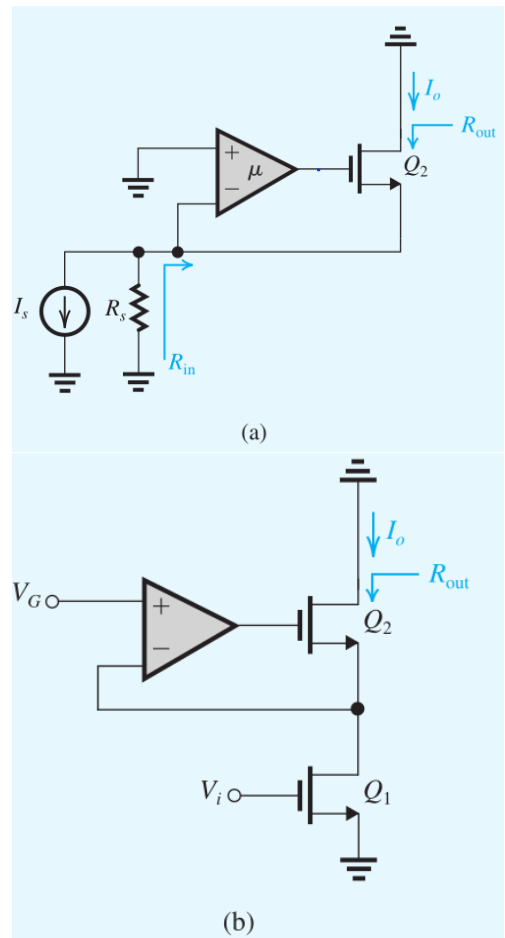
A-circuit



Question 4

The feedback current amplifier in figure (a) can be thought of as a “super” CG transistor. Note that rather than connecting the gate of Q_2 to signal ground, an amplifier is placed between source and gate.

- If μ is very large, what is the signal voltage at the input terminal? What is the input resistance? What is the current gain I_o/I_s ?
- For finite μ but assuming that the input resistance of the amplifier μ is very large, find the A circuit and derive expressions for A, R_i , and R_o .
- What is the value of β ?
- Find $A\beta$ and A_f . If μ is large, what is the value of A_f ?
- Find R_{in} and R_{out} assuming the loop gain is large.
- The “super” CG transistor can be utilized in the cascode configuration shown in figure (b), where V_G is a DC bias voltage. Replacing Q_1 by its small-signal model, use the analogy of the resulting circuit to that in figure (a) to find I_o and R_{out} .



Soln:

$$(a) \quad V_{G_2} = \mu(V_+ - V_-)$$

$$\therefore V_+ - V_- = \frac{V_{G_2}}{\mu} \quad \because \mu \text{ is very large} \Rightarrow V_+ - V_- \approx 0 \quad \therefore V_+ \approx V_-$$

$$V_- = V_+ = 0 \quad \therefore \text{Signal voltage at the input terminal} = 0V.$$

$\therefore$ input resistance is also zero.

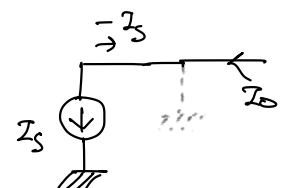
As voltage drop across $R_s \approx 0$; no current will flow through it.

$$I_s \approx I_o$$

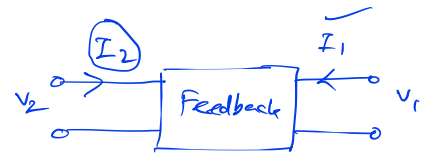
$$\therefore \frac{I_o}{I_s} \approx 1.$$

$$-I_s + I_o = 0$$

$$\therefore I_s = I_o$$

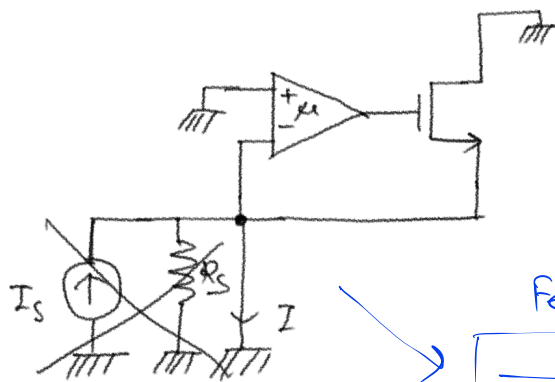


(b) This is a current amplifier

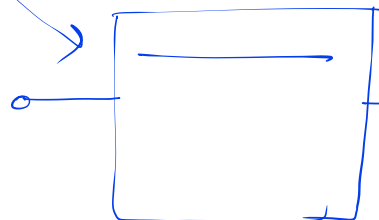


$$V_1 = -I_1 R + V_2$$

$$I_2 = \beta I_1 + V_2$$

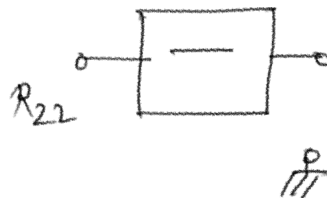
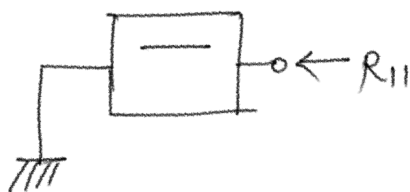


Feedback Network

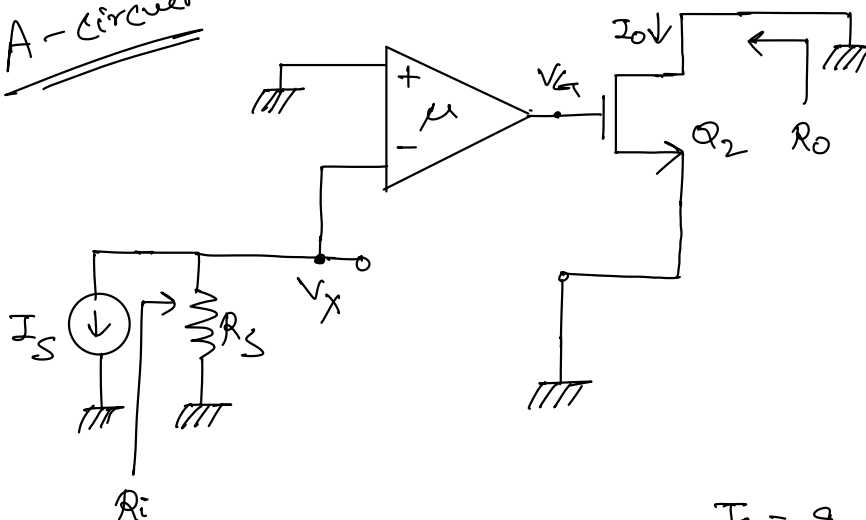


$$\beta = \frac{I_2}{I_1} \bigg|_{V_2=0}$$

$$R_{11} = \frac{V_1}{I_1} \bigg|_{V_2=0} \quad G_{22} = \frac{I_2}{V_2} \bigg|_{I_1=0}$$



A-circuit



$$V_X = -I_S R_S$$

$$V_G = \mu(V_+ - V_-)$$

$$= \mu(0 + V_X)$$

$$= +\mu V_X$$

$$= \mu I_S R_S$$

$$I_O = g_{m2} V_{GS} = g_{m2} \mu I_S R_S$$

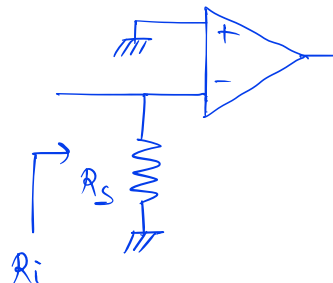
$$\text{Gain } A = \frac{I_O}{I_S} = \mu g_{m2} R_S$$

Our regular current amplifier, the input current goes into the input side, but here the dirⁿ is opposite. ; so for gain calculation, I_S should be replaced by $-I_S$.

$$\text{In that case, } A = \frac{I_O}{(-I_S)} = -\mu g_{m2} R_S$$

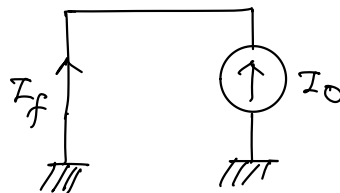
$$\underline{R_i} \quad R_i = R_S$$

$$R_o = r_{o2}$$



$$(c) \quad I_f = -I_o$$

$$\boxed{\frac{I_o}{I_f} = -1 = \beta}$$



$$\beta = \frac{I_o}{I_f}$$

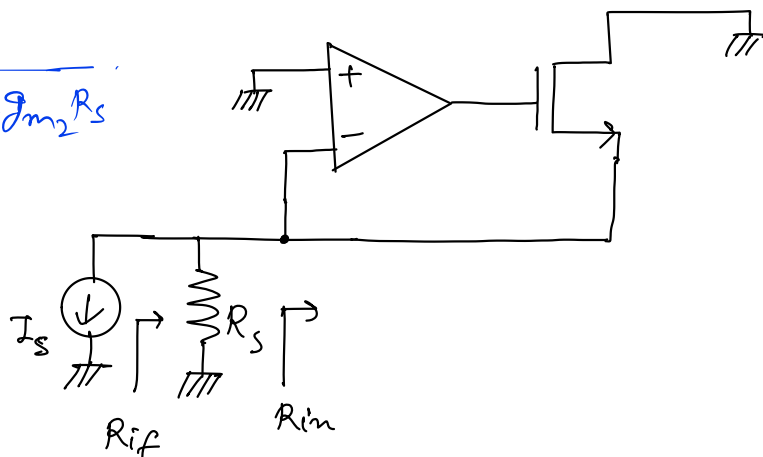
$$(d) \quad A\beta = (-\mu g_{m2} R_S)(-1) = \mu g_{m2} R_S.$$

$$A_f = \frac{A}{1+A\beta} = \frac{-\mu g_{m2} R_S}{1 + \mu g_{m2} R_S} \approx -1 \quad [\text{as } \mu \text{ is large}]$$

$$(e) \quad R_{if} = \frac{R_i}{1+A\beta} = \frac{R_S}{1 + \mu g_{m2} R_S}$$

$$\frac{1}{R_{if}} = \frac{1}{R_S} + \frac{1}{R_{in}}$$

$$\frac{1}{R_{in}} = \frac{1}{R_{if}} - \frac{1}{R_S}$$



$$R_{in} = \frac{1}{\frac{1}{R_{if}} - \frac{1}{R_S}} = \frac{1}{\frac{1 + \mu g_{m2} R_S}{R_S} - \frac{1}{R_S}} = \frac{R_S}{1 + \mu g_{m2} R_S - 1}$$

$$\boxed{R_{in} = \frac{1}{\mu g_{m2}}}$$

$$R_{of} = R_o(1+A\beta) = r_{o2}(1+A\beta) = r_{o2}(1 + \mu g_{m2} R_S) = R_{out}.$$

(f). As voltage across $r_o \approx 0$

$$i_o = g_{m1} V_i$$

Using previous formula

$$R_{out} = r_{o2} (1 + \mu g_{m2} r_{o1})$$

$$= r_{o2} + \mu g_{m2} r_{o1} r_{o2}$$

$$\underline{R_{out} \approx \mu g_{m2} r_{o1} r_{o2}}$$

